able. This is the *Bayesian* approach, which supposes that prior to observing the data, information about θ is always available to the decision maker, and that this information can be expressed in terms of a probability distribution on θ. In such a situation, we show how to compute the *Bayes estimator*, which is the estimator whose expected squared distance from θ is minimal.

7.2 Maximum likelihood estimators

Any statistic used to estimate the value of an unknown parameter θ is called an *estimator* of θ. The observed value of the estimator is called the *estimate*. For instance, as we shall see, the usual estimator of the mean of a normal population, based on a sample $X_1, \ldots, X_n$ from that population, is the sample mean $\overline{X} = \sum_i X_i/n$. If a sample of size 3 yields the data $X_1 = 2$, $X_2 = 3$, $X_3 = 4$, then the estimate of the population mean, resulting from the estimator $\overline{X}$, is the value 3.

Suppose that the random variables $X_1, \ldots, X_n$, whose joint distribution is assumed given except for an unknown parameter θ, are to be observed. The problem of interest is to use the observed values to estimate θ. For example, the X_i's might be independent, exponential random variables each having the same unknown mean θ. In this case, the joint density function of the random variables would be given by

$$
\begin{aligned}
&f(x_1, x_2, \ldots, x_n) \\
&\quad = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n) \\
&\quad = \frac{1}{\theta} e^{-x_1/\theta} \frac{1}{\theta} e^{-x_2/\theta} \cdots \frac{1}{\theta} e^{-x_n/\theta}, \qquad 0 < x_i < \infty, i = 1, \ldots, n \\
&\quad = \frac{1}{\theta^n} \exp\left\{ -\sum_1^n x_i/\theta \right\}, \qquad 0 < x_i < \infty, i = 1, \ldots, n
\end{aligned}
$$

and the objective would be to estimate θ from the observed data $X_1, X_2, \ldots, X_n$.

A particular type of estimator, known as the *maximum likelihood* estimator, is widely used in statistics. It is obtained by reasoning as follows. Let $f(x_1, \ldots, x_n|\theta)$ denote the joint probability mass function of the random variables $X_1, X_2, \ldots, X_n$ when they are discrete, and let it be their joint probability density function when they are jointly continuous random variables. Because θ is assumed unknown, we also write f as a function of θ. Now since $f(x_1, \ldots, x_n|\theta)$ represents the likelihood that the values $x_1, x_2, \ldots, x_n$ will be observed when θ is the true value of the parameter, it would seem that a reasonable estimate of θ would be that value yielding the largest likelihood of the observed values. In other words, the maximum likelihood estimate $\hat{\theta}$ is defined to be that value of θ maximizing $f(x_1, \ldots, x_n|\theta)$ where $x_1, \ldots, x_n$ are the ob-

served values. The function $f(x_1, \ldots, x_n|\theta)$ is often referred to as the *likelihood* function of θ.

In determining the maximizing value of θ, it is often useful to use the fact that $f(x_1, \ldots, x_n|\theta)$ and $\log[f(x_1, \ldots, x_n|\theta)]$ have their maximum at the same value of θ. Hence, we may also obtain $\hat{\theta}$ by maximizing $\log[f(x_1, \ldots, x_n|\theta)]$.

Example 7.2.a (Maximum Likelihood Estimator of a Bernoulli Parameter). Suppose that n independent trials, each of which is a success with probability p, are performed. What is the maximum likelihood estimator of p?

Solution. The data consist of the values of $X_1, \ldots, X_n$ where

$$X_i = \begin{cases} 1 & \text{if trial } i \text{ is a success} \\ 0 & \text{otherwise} \end{cases}$$

Now

$$P\{X_i = 1\} = p = 1 - P\{X_i = 0\}$$

which can be succinctly expressed as

$$P\{X_i = x\} = p^x(1-p)^{1-x}, \quad x = 0, 1$$

Hence, by the assumed independence of the trials, the likelihood (that is, the joint probability mass function) of the data is given by

$$\begin{aligned} f(x_1, \ldots, x_n|p) &= P\{X_1 = x_1, \ldots, X_n = x_n|p\} \\ &= p^{x_1}(1-p)^{1-x_1} \cdots p^{x_n}(1-p)^{1-x_n} \\ &= p^{\sum_1^n x_i}(1-p)^{n-\sum_1^n x_i}, \quad x_i = 0, 1, \quad i = 1, \ldots, n \end{aligned}$$

To determine the value of p that maximizes the likelihood, first take logs to obtain

$$\log f(x_1, \ldots, x_n|p) = \sum_1^n x_i \log p + \left(n - \sum_1^n x_i\right) \log(1-p)$$

Differentiation yields

$$\frac{d}{dp} \log f(x_1, \ldots, x_n|p) = \frac{\sum_1^n x_i}{p} - \frac{\left(n - \sum_1^n x_i\right)}{1-p}$$

Upon equating to zero and solving, we obtain that the maximum likelihood estimate $\hat{p}$ satisfies

$$\frac{\sum_1^n x_i}{\hat{p}} = \frac{n - \sum_1^n x_i}{1 - \hat{p}}$$

or

$$\hat{p} = \frac{\sum_{i=1}^n x_i}{n}$$

Hence, the maximum likelihood estimator of the unknown mean of a Bernoulli distribution is given by

$$d(X_1, \ldots, X_n) = \frac{\sum_{i=1}^n X_i}{n}$$

Since $\sum_{i=1}^n X_i$ is the number of successful trials, we see that the maximum likelihood estimator of p is equal to the proportion of the observed trials that result in successes. For an illustration, suppose that each RAM (random access memory) chip produced by a certain manufacturer is, independently, of acceptable quality with probability p. Then if out of a sample of 1000 tested 921 are acceptable, it follows that the maximum likelihood estimate of p is .921. ■

Example 7.2.b. Two proofreaders were given the same manuscript to read. If proofreader 1 found n_1 errors, and proofreader 2 found n_2 errors, with $n_{1,2}$ of these errors being found by both proofreaders, estimate N, the total number of errors that are in the manuscript.

Solution. Before we can estimate N we need to make some assumptions about the underlying probability model. So let us assume that the results of the proofreaders are independent, and that each error in the manuscript is independently found by proofreader i with probability p_i, $i = 1, 2$.

To estimate N, we will start by deriving an estimator of p_1. To do so, note that each of the n_2 errors found by reader 2 will, independently, be found by proofreader 1 with probability p_i. Because proofreader 1 found $n_{1,2}$ of those n_2 errors, a reasonable estimate of p_1 is given by

$$\hat{p}_1 = \frac{n_{1,2}}{n_2}$$

However, because proofreader 1 found n_1 of the N errors in the manuscript, it is reasonable to suppose that p_1 is also approximately equal to $\frac{n_1}{N}$. Equating

this to $\hat{p}_1$ gives that

$$\frac{n_{1,2}}{n_2} \approx \frac{n_1}{N}$$

or

$$N \approx \frac{n_1 n_2}{n_{1,2}}$$

Because the preceding estimate is symmetric in n_1 and n_2, it follows that it is the same no matter which proofreader is designated as proofreader 1.

An interesting application of the preceding occurred when two teams of researchers recently announced that they had decoded the human genetic code sequence. As part of their work both teams estimated that the human genome consisted of approximately 33,000 genes. Because both teams independently arrived at the same number, many scientists found this number believable. However, most scientists were quite surprised by this relatively small number of genes; by comparison it is only about twice as many as a fruit fly has. However, a closer inspection of the findings indicated that the two groups only agreed on the existence of about 17,000 genes. (That is, 17,000 genes were found by both teams.) Thus, based on our preceding estimator, we would estimate that the actual number of genes, rather than being 33,000, is

$$\frac{n_1 n_2}{n_{1,2}} = \frac{33{,}000 \times 33{,}000}{17{,}000} \approx 64{,}000$$

(Because there is some controversy about whether some of the genes claimed to be found are actually genes, 64,000 should probably be taken as an upper bound on the actual number of genes.)

The estimation approach used when there are two proofreaders does not work when there are m proofreaders, when $m > 2$. Because, if for each i, we let $\hat{p}_i$ be the fraction of the errors found by at least one of the other proofreaders j, $(j \neq i)$, that are also found by i, and then set that equal to $\frac{n_i}{N}$, then the estimate of N, namely $\frac{n_i}{\hat{p}_i}$, would differ for different values of i. Moreover, with this approach it is possible that we may have that $\hat{p}_i > \hat{p}_j$ even if proofreader i finds fewer errors than does proofreader j. For instance, for $m = 3$, suppose proofreaders 1 and 2 find exactly the same set of 10 errors whereas proofreader 3 finds 20 errors with only 1 of them in common with the set of errors found by the others. Then, because proofreader 1 (and 2) found 10 of the 29 errors found by at least one of the other proofreaders, $\hat{p}_i = 10/29$, $i = 1, 2$. On the other hand, because proofreader 3 only found 1 of the 10 errors found by the others, $\hat{p}_3 = 1/10$. Therefore, although proofreader 3 found twice the number of errors as did proofreader 1, the estimate of p_3 is less than that of p_1. To obtain more reasonable estimates, we could take the preceding values of $\hat{p}_i$, $i = 1, \ldots, m$, as preliminary estimates of the p_i. Now, let n_f be the number of errors that

are found by at least one proofreader. Because n_f/N is the fraction of errors that are found by at least one proofreader, this should approximately equal $1 - \prod_{i=1}^{m}(1-p_i)$, the probability that an error is found by at least one proofreader. Therefore, we have

$$\frac{n_f}{N} \approx 1 - \prod_{i=1}^{m}(1-p_i)$$

suggesting that $N \approx \hat{N}$, where

$$\hat{N} = \frac{n_f}{1 - \prod_{i=1}^{m}(1-\hat{p}_i)} \tag{7.2.1}$$

With this estimate of N, we can then reset our estimates of the p_i by using

$$\hat{p}_i = \frac{n_i}{\hat{N}}, \quad i = 1, \ldots, m \tag{7.2.2}$$

We can then reestimate N by using the new value (Equation (7.2.1)). (The estimation need not stop here; each time we obtain a new estimate $\hat{N}$ of N we can use Equation (7.2.2) to obtain new estimates of the p_i, which can then be used to obtain a new estimate of N, and so on.) ■

Example 7.2.c (Maximum Likelihood Estimator of a Poisson Parameter). Suppose $X_1, \ldots, X_n$ are independent Poisson random variables each having mean λ. Determine the maximum likelihood estimator of λ.

Solution. The likelihood function is given by

$$\begin{aligned} f(x_1, \ldots, x_n|\lambda) &= \frac{e^{-\lambda}\lambda^{x_1}}{x_1!} \cdots \frac{e^{-\lambda}\lambda^{x_n}}{x_n!} \\ &= \frac{e^{-n\lambda}\lambda^{\sum_1^n x_i}}{x_1! \ldots x_n!} \end{aligned}$$

Thus,

$$\log\, f(x_1, \ldots, x_n|\lambda) = -n\lambda + \sum_{1}^{n} x_i \log\lambda - \log c$$

where $c = \prod_{i=1}^{n} x_i!$ does not depend on λ. Differentiation yields

$$\frac{d}{d\lambda} \log\, f(x_1, \ldots, x_n|\lambda) = -n + \frac{\sum_{1}^{n} x_i}{\lambda}$$

By equating to zero, we obtain that the maximum likelihood estimate $\hat{\lambda}$ equals

$$\hat{\lambda} = \frac{\sum_1^n x_i}{n}$$

and so the maximum likelihood estimator is given by

$$d(X_1, \ldots, X_n) = \frac{\sum_{i=1}^n X_i}{n}$$

For example, suppose that the number of people who enter a certain retail establishment in any day is a Poisson random variable having an unknown mean λ, which must be estimated. If after 20 days a total of 857 people have entered the establishment, then the maximum likelihood estimate of λ is $857/20 = 42.85$. That is, we estimate that on average, 42.85 customers will enter the establishment on a given day. ■

Example 7.2.d. The number of traffic accidents in Berkeley, California, in 10 randomly chosen nonrainy days in 1998 is as follows:

4, 0, 6, 5, 2, 1, 2, 0, 4, 3

Use these data to estimate the proportion of nonrainy days that had 2 or fewer accidents that year.

Solution. Since there are a large number of drivers, each of whom has a small probability of being involved in an accident in a given day, it seems reasonable to assume that the daily number of traffic accidents is a Poisson random variable. Since

$$\overline{X} = \frac{1}{10}\sum_{i=1}^{10} X_i = 2.7$$

it follows that the maximum likelihood estimate of the Poisson mean is 2.7. Since the long-run proportion of nonrainy days that have 2 or fewer accidents is equal to $P\{X \leq 2\}$, where X is the random number of accidents in a day, it follows that the desired estimate is

$$e^{-2.7}(1 + 2.7 + (2.7)^2/2) = .4936$$

That is, we estimate that a little less than half of the nonrainy days had 2 or fewer accidents. ■

Example 7.2.e (Maximum Likelihood Estimator in a Normal Population). Suppose $X_1, \ldots, X_n$ are independent, normal random variables each with unknown mean μ and unknown standard deviation σ. The joint density is given

by

$$f(x_1, \ldots, x_n|\mu, \sigma) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[\frac{-(x_i - \mu)^2}{2\sigma^2}\right]$$

$$= \left(\frac{1}{2\pi}\right)^{n/2} \frac{1}{\sigma^n} \exp\left[\frac{-\sum_1^n (x_i - \mu)^2}{2\sigma^2}\right]$$

The logarithm of the likelihood is thus given by

$$\log\, f(x_1, \ldots, x_n|\mu, \sigma) = -\frac{n}{2}\log(2\pi) - n\log\sigma - \frac{\sum_1^n (x_i - \mu)^2}{2\sigma^2}$$

In order to find the value of μ and σ maximizing the foregoing, we compute

$$\frac{\partial}{\partial\mu} \log\, f(x_1, \ldots, x_n|\mu, \sigma) = \frac{\sum_{i=1}^n (x_i - \mu)}{\sigma^2}$$

$$\frac{\partial}{\partial\sigma} \log\, f(x_1, \ldots, x_n|\mu, \sigma) = -\frac{n}{\sigma} + \frac{\sum_1^n (x_i - \mu)^2}{\sigma^3}$$

Equating these equations to zero yields that

$$\hat{\mu} = \sum_{i=1}^{n} x_i/n$$

and

$$\hat{\sigma} = \left[\sum_{i=1}^{n} (x_i - \hat{\mu})^2/n\right]^{1/2}$$

Hence, the maximum likelihood estimators of μ and σ are given, respectively, by

$$\overline{X} \quad \text{and} \quad \left[\sum_{i=1}^{n} (X_i - \overline{X})^2/n\right]^{1/2} \tag{7.2.3}$$

It should be noted that the maximum likelihood estimator of the standard deviation σ differs from the sample standard deviation

$$S = \left[\sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1) \right]^{1/2}$$

in that the denominator in Equation (7.2.3) is $\sqrt{n}$ rather than $\sqrt{n-1}$. However, for n of reasonable size, these two estimators of σ will be approximately equal. ■

Example 7.2.f. *Kolmogorov's law of fragmentation* states that the size of an individual particle in a large collection of particles resulting from the fragmentation of a mineral compound will have an approximate lognormal distribution, where a random variable X is said to have a *lognormal* distribution if $\log(X)$ has a normal distribution. The law, which was first noted empirically and then later given a theoretical basis by Kolmogorov, has been applied to a variety of engineering studies. For instance, it has been used in the analysis of the size of randomly chosen gold particles from a collection of gold sand. A less obvious application of the law has been to a study of the stress release in earthquake fault zones (see Lomnitz, C., "Global Tectonics and Earthquake Risk," *Developments in Geotectonics*, Elsevier, Amsterdam, 1979).

Suppose that a sample of 10 grains of metallic sand taken from a large sand pile have respective lengths (in millimeters):

2.2, 3.4, 1.6, 0.8, 2.7, 3.3, 1.6, 2.8, 2.5, 1.9

Estimate the percentage of sand grains in the entire pile whose length is between 2 and 3 mm.

Solution. Taking the natural logarithm of these 10 data values, the following transformed data set results

.7885, 1.2238, .4700, −.2231, .9933, 1.1939, .4700, 1.0296, .9163, .6419

Because the sample mean and sample standard deviation of these data are

$$\overline{x} = .7504, \quad s = .4351$$

it follows that the logarithm of the length of a randomly chosen grain has a normal distribution with mean approximately equal to .7504 and with standard deviation approximately equal to .4351. Hence, if X is the length of the grain, then

$$\begin{aligned} P\{2 < X < 3\} &= P\{\log(2) < \log(X) < \log(3)\} \\ &= P\left\{ \frac{\log(2) - .7504}{.4351} < \frac{\log(X) - .7504}{.4351} < \frac{\log(3) - .7504}{.4351} \right\} \end{aligned}$$

$$\begin{aligned}
&\approx \text{pnorm}((\log(3) - .7504)/.4351) \\
&\quad - \text{pnorm}((\log(2) - .7504)/.4351) \\
&= 0.3405766 \quad \blacksquare
\end{aligned}$$

The lognormal distribution is often assumed in situations where the random variable under interest can be regarded as the product of a large number of independent and identically distributed random variables. For instance, it is commonly used in finance as the distribution of the price of a security at some future time. To see why this might be reasonable, suppose that the current price of the security is s and that we are interested in $S(t)$, the price of the security after an additional time t. For a large value n, let $t_i = it/n$, and consider $S(t_1), \ldots, S(t_n)$, the prices of the security at the times $t_1, \ldots, t_n$. Now, a common assumption in finance is that the ratios $S(t_i)/S(t_{i-1})$ are approximately independent and identically distributed. Consequently, if we let $X_i = S(t_i)/S(t_{i-1})$, then writing

$$\begin{aligned}
S(t) = S(t_n) &= S(t_0) \cdot \frac{S(t_1)}{S(t_0)} \cdot \frac{S(t_2)}{S(t_1)} \cdots \frac{S(t_n)}{S(t_{n-1})} \\
&= s \prod_{i=1}^{n} X_i
\end{aligned}$$

we obtain, upon taking logarithms, that

$$\log(S(t)) = \log(s) + \sum_{i=1}^{n} \log(X_i)$$

Thus, by the central limit theorem $\log(S(t))$ will approximately have a normal distribution.

The lognormal distribution has also been shown to be a good fit for such random variables as length of patient stays in hospitals, and vehicle travel times.

In all of the foregoing examples, the maximum likelihood estimator of the population mean turned out to be the sample mean $\overline{X}$. To show that this is not always the situation, consider the following example.

Example 7.2.g (Estimating the Mean of a Uniform Distribution). Suppose $X_1, \ldots, X_n$ constitute a sample from a uniform distribution on $(0, \theta)$, where θ is unknown. Their joint density is thus

$$f(x_1, x_2, \ldots, x_n|\theta) = \begin{cases} \dfrac{1}{\theta^n} & 0 < x_i < \theta, \quad i = 1, \ldots, n \\ 0 & \text{otherwise} \end{cases}$$

This density is maximized by choosing θ as small as possible. Since θ must be at least as large as all of the observed values x_i, it follows that the smallest possible

choice of θ is equal to $\max(x_1, x_2, \ldots, x_n)$. Hence, the maximum likelihood estimator of θ is

$$\hat{\theta} = \max(X_1, X_2, \ldots, X_n)$$

It easily follows from the foregoing that the maximum likelihood estimator of $\theta/2$, the mean of the distribution, is $\max(X_1, X_2, \ldots, X_n)/2$. ■

7.2.1 Estimating life distributions[1]

Let X denote the age at death of a randomly chosen child born today. That is, $X = i$ if the newborn dies in its ith year, $i \geq 1$. To estimate the probability mass function of X, let λ_i denote the probability that a newborn who has survived his or her first $i - 1$ years dies in year i. That is,

$$\lambda_i = P\{X = i | X > i - 1\} = \frac{P\{X = i\}}{P\{X > i - 1\}}$$

Also, let

$$s_i = 1 - \lambda_i = \frac{P\{X > i\}}{P\{X > i - 1\}}$$

be the probability that a newborn who survives her first $i - 1$ years also survives year i. The quantity λ_i is called the *failure rate*, and s_i is called the *survival rate*, of an individual who is entering his or her ith year. Now,

$$\begin{aligned} s_1 s_2 \cdots s_i &= P\{X > 1\} \frac{P\{X > 2\} P\{X > 3\}}{P\{X > 1\} P\{X > 2\}} \cdots \frac{P\{X > i\}}{P\{X > i - 1\}} \\ &= P\{X > i\} \end{aligned}$$

Therefore,

$$P\{X = n\} = P\{X > n - 1\} \lambda_n = s_1 \cdots s_{n-1}(1 - s_n)$$

Consequently, we can estimate the probability mass function of X by estimating the quantities s_i, $i = 1, \ldots, n$. The value s_i can be estimated by looking at all individuals in the population who reached age i 1 year ago, and then letting the estimate $\hat{s}_i$ be the fraction of them who are alive today. We would then use $\hat{s}_1 \hat{s}_2 \cdots \hat{s}_{n-1}(1 - \hat{s}_n)$ as the estimate of $P\{X = n\}$. (Note that although we are using the most recent possible data to estimate the quantities s_i, our estimate of the probability mass function of the lifetime of a newborn assumes that the survival rate of the newborn when it reaches age i will be the same as last year's survival rate of someone of age i.)

[1] Optional section.

The use of the survival rate to estimate a life distribution is also of importance in health studies with partial information. For instance, consider a study in which a new drug is given to a random sample of 12 lung cancer patients. Suppose that after some time we have the following data on the number of months of survival after starting the new drug:

$$4, 7^*, 9, 11^*, 12, 3, 14^*, 1, 8, 7, 5, 3^*$$

where x means that the patient died in month x after starting the drug treatment, and x^* means that the patient has taken the drug for x months and is still alive.

Let X equal the number of months of survival after beginning the drug treatment, and let

$$s_i = P\{X > i | X > i - 1\} = \frac{P\{X > i\}}{P\{X > i - 1\}}$$

To estimate s_i, the probability that a patient who has survived the first $i - 1$ months will also survive month i, we should take the fraction of those patients who began their ith month of drug taking and survived the month. For instance, because 11 of the 12 patients survived month 1, $\hat{s}_1 = 11/12$. Because all 11 patients who began month 2 survived, $\hat{s}_2 = 11/11$. Because 10 of the 11 patients who began month 3 survived, $\hat{s}_3 = 10/11$. Because 8 of the 9 patients who began their fourth month of taking the drug (the 9 being all but the ones labeled 1, 3, and 3*) survived month 4, $\hat{s}_4 = 8/9$. Similar reasoning holds for the others, giving the following survival rate estimates:

$$\hat{s}_1 = 11/12$$
$$\hat{s}_2 = 11/11$$
$$\hat{s}_3 = 10/11$$
$$\hat{s}_4 = 8/9$$
$$\hat{s}_5 = 7/8$$
$$\hat{s}_6 = 7/7$$
$$\hat{s}_7 = 6/7$$
$$\hat{s}_8 = 4/5$$
$$\hat{s}_9 = 3/4$$
$$\hat{s}_{10} = 3/3$$
$$\hat{s}_{11} = 3/3$$
$$\hat{s}_{12} = 1/2$$
$$\hat{s}_{13} = 1/1$$
$$\hat{s}_{14} = 1/1$$

We can now use $\prod_{i=1}^{j} \hat{s}_i$ to estimate the probability that a drug taker survives at least j time periods, $j = 1, \ldots, 14$. For instance, our estimate of $P\{X > 6\}$ is 35/54.

7.3 Interval estimates

Suppose that $X_1, \ldots, X_n$ is a sample from a normal population having unknown mean μ and known variance σ^2. It has been shown that $\overline{X} = \sum_{i=1}^{n} X_i/n$ is the maximum likelihood estimator for μ. However, we don't expect that the sample mean $\overline{X}$ will exactly equal μ, but rather that it will "be close." Hence, rather than a point estimate, it is sometimes more valuable to be able to specify an interval for which we have a certain degree of confidence that μ lies within. To obtain such an interval estimator, we make use of the probability distribution of the point estimator. Let us see how it works for the preceding situation.

In the foregoing, since the point estimator $\overline{X}$ is normal with mean μ and variance σ^2/n, it follows that

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \sqrt{n}\frac{(\overline{X} - \mu)}{\sigma}$$

has a standard normal distribution. Therefore,

$$P\left\{-1.96 < \sqrt{n}\frac{(\overline{X} - \mu)}{\sigma} < 1.96\right\} = .95$$

or, equivalently,

$$P\left\{-1.96\frac{\sigma}{\sqrt{n}} < \overline{X} - \mu < 1.96\frac{\sigma}{\sqrt{n}}\right\} = .95$$

Multiplying through by -1 yields the equivalent statement

$$P\left\{-1.96\frac{\sigma}{\sqrt{n}} < \mu - \overline{X} < 1.96\frac{\sigma}{\sqrt{n}}\right\} = .95$$

or, equivalently,

$$P\left\{\overline{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right\} = .95$$

That is, 95 percent of the time the value of the sample average $\bar{X}$ will be such that the distance between it and the mean μ will be less than $1.96\sigma/\sqrt{n}$. If we now observe the sample and it turns out that $\overline{X} = \overline{x}$, then we say that "with 95 percent confidence"

$$\overline{x} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \overline{x} + 1.96\frac{\sigma}{\sqrt{n}} \tag{7.3.1}$$

That is, "with 95 percent confidence" we assert that the true mean lies within $1.96\sigma/\sqrt{n}$ of the observed sample mean. The interval

$$\left(\overline{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

is called a *95 percent confidence interval estimate* of μ.

Example 7.3.a. Suppose that when a signal having value μ is transmitted from location A the value received at location B is normally distributed with mean μ and variance 4. That is, if μ is sent, then the value received is $\mu + N$ where N, representing noise, is normal with mean 0 and variance 4. To reduce error, suppose the same value is sent 9 times. If the successive values received are 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, let us construct a 95 percent confidence interval for μ.

Since

$$\overline{x} = \frac{81}{9} = 9$$

it follows, under the assumption that the values received are independent, that a 95 percent confidence interval for μ is

$$\left(9 - 1.96\frac{\sigma}{3},\ 9 + 1.96\frac{\sigma}{3}\right) = (7.69, 10.31)$$

Hence, we are "95 percent confident" that the true message value lies between 7.69 and 10.31. ■

The interval in Equation (7.3.1) is called a *two-sided confidence interval*. Sometimes, however, we are interested in determining a value so that we can assert with, say, 95 percent confidence, that μ is at least as large as that value.

To determine such a value, note that if Z is a standard normal random variable then

$$P\{Z < 1.645\} = .95$$

As a result,

$$P\left\{\sqrt{n}\frac{(\overline{X} - \mu)}{\sigma} < 1.645\right\} = .95$$

or

$$P\left\{\overline{X} - 1.645\frac{\sigma}{\sqrt{n}} < \mu\right\} = .95$$

Thus, a 95 percent *one-sided upper confidence interval* for μ is

$$\left(\overline{x} - 1.645\frac{\sigma}{\sqrt{n}}, \infty\right)$$

where $\overline{x}$ is the observed value of the sample mean.

A *one-sided lower confidence interval* is obtained similarly; when the observed value of the sample mean is $\overline{x}$, then the 95 percent one-sided lower confidence interval for μ is

$$\left(-\infty, \overline{x} + 1.645\frac{\sigma}{\sqrt{n}}\right)$$

Example 7.3.b. Determine the upper and lower 95 percent confidence interval estimates of μ in Example 7.3.a.

Solution. Since

$$1.645\frac{\sigma}{\sqrt{n}} = \frac{3.29}{3} = 1.097$$

the 95 percent upper confidence interval is

$$(9 - 1.097, \infty) = (7.903, \infty)$$

and the 95 percent lower confidence interval is

$$(-\infty, 9 + 1.097) = (-\infty, 10.097) \quad \blacksquare$$

We can also obtain confidence intervals of any specified level of confidence. To do so, recall that z_α is such that

$$P\{Z > z_\alpha\} = \alpha$$

when Z is a standard normal random variable. But this implies (see Figure 7.1) that for any α

$$P\{-z_{\alpha/2} < Z < z_{\alpha/2}\} = 1 - \alpha$$

As a result, we see that

$$P\left\{-z_{\alpha/2} < \sqrt{n}\frac{(\overline{X} - \mu)}{\sigma} < z_{\alpha/2}\right\} = 1 - \alpha$$

or

$$P\left\{-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \overline{X} - \mu < z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

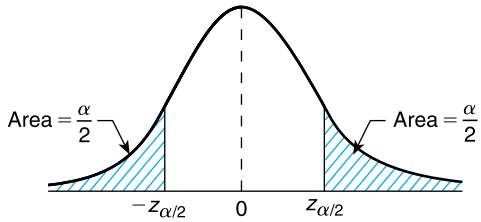


FIGURE 7.1
$P\{-z_{\alpha/2} < Z < z_{\alpha/2}\} = 1 - \alpha$.

or

$$P\left\{-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu - \overline{X} < z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

That is,

$$P\left\{\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

Hence, a $100(1 - \alpha)$ percent two-sided confidence interval for μ is

$$\left(\overline{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \overline{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

where $\overline{x}$ is the observed sample mean.

Similarly, knowing that $Z = \sqrt{n}\frac{(\overline{X} - \mu)}{\sigma}$ is a standard normal random variable, along with the identities

$$P\{Z > z_\alpha\} = \alpha$$

and

$$P\{Z < -z_\alpha\} = \alpha$$

results in one-sided confidence intervals of any desired level of confidence. Specifically, we obtain that

$$\left(\overline{x} - z_\alpha\frac{\sigma}{\sqrt{n}}, \ \infty\right)$$

and

$$\left(-\infty, \ \overline{x} + z_\alpha\frac{\sigma}{\sqrt{n}}\right)$$

are, respectively, $100(1 - \alpha)$ percent one-sided upper and $100(1 - \alpha)$ percent one-sided lower confidence intervals for μ.

Example 7.3.c. Use the data of Example 7.3.a to obtain a 99 percent confidence interval estimate of μ, along with 99 percent one-sided upper and lower intervals.

Solution. Since $z_{.005} = 2.58$, and

$$2.58\frac{\alpha}{\sqrt{n}} = \frac{5.16}{3} = 1.72$$

it follows that a 99 percent confidence interval for μ is

$$9 \pm 1.72$$

That is, the 99 percent confidence interval estimate is $(7.28, 10.72)$.

Also, since $z_{.01} = 2.33$, a 99 percent upper confidence interval is

$$(9 - 2.33(2/3),\ \infty) = (7.447,\ \infty)$$

Similarly, a 99 percent lower confidence interval is

$$(-\infty,\ 9 + 2.33(2/3)) = (-\infty,\ 10.553) \quad \blacksquare$$

Sometimes we are interested in a two-sided confidence interval of a certain level, say $1 - \alpha$, and the problem is to choose the sample size n so that the interval is of a certain size. For instance, suppose that we want to compute an interval of length .1 that we can assert, with 99 percent confidence, contains μ. How large need n be? To solve this, note that as $z_{.005} = 2.58$ it follows that the 99 percent confidence interval for μ from a sample of size n is

$$\left(\overline{x} - 2.58\frac{\sigma}{\sqrt{n}},\quad \overline{x} + 2.58\frac{\sigma}{\sqrt{n}}\right)$$

Hence, its length is

$$5.16\frac{\sigma}{\sqrt{n}}$$

Thus, to make the length of the interval equal to .1, we must choose

$$5.16\frac{\sigma}{\sqrt{n}} = .1$$

or

$$n = (51.6\sigma)^2$$

Remark

The interpretation of "a $100(1-\alpha)$ percent confidence interval" can be confusing. It should be noted that we are *not* asserting that the probability that $\mu \in (\overline{x} - 1.96\sigma/\sqrt{n},\ \overline{x} + 1.96\sigma/\sqrt{n})$ is .95, for there are no random variables involved in this assertion. What we are asserting is that the technique utilized to obtain this interval is such that 95 percent of the time that it is employed it will result in an interval in which μ lies. In other words, before the data are observed we can assert that with probability .95 the interval that will be obtained will contain μ, whereas after the data are obtained we can only assert that the resultant interval indeed contains μ "with confidence .95."

Example 7.3.d. From past experience it is known that the weights of salmon grown at a commercial hatchery are normal with a mean that varies from season to season but with a standard deviation that remains fixed at 0.3 pound. If we want to be 95 percent certain that our estimate of the present season's mean weight of a salmon is correct to within ± 0.1 pound, how large a sample is needed?

Solution. A 95 percent confidence interval estimate for the unknown mean μ, based on a sample of size n, is

$$\mu \in \left(\overline{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

Because the estimate $\overline{x}$ is within $1.96(\sigma/\sqrt{n}) = .588/\sqrt{n}$ of any point in the interval, it follows that we can be 95 percent certain that $\overline{x}$ is within 0.1 of μ provided that

$$\frac{.588}{\sqrt{n}} \leq 0.1$$

That is, provided that

$$\sqrt{n} \geq 5.88$$

or

$$n \geq 34.57$$

That is, a sample size of 35 or larger will suffice. ■

7.3.1 Confidence interval for a normal mean when the variance is unknown

Suppose now that $X_1, \ldots, X_n$ is a sample from a normal distribution with unknown mean μ and unknown variance σ^2, and that we wish to construct a

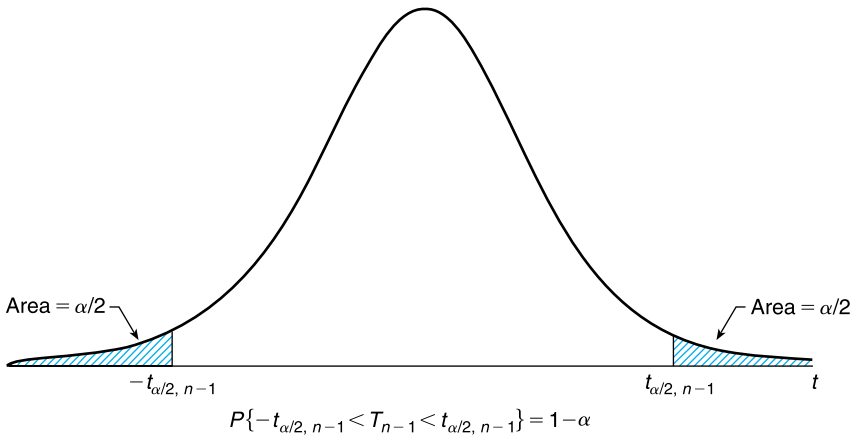


FIGURE 7.2
t-density function.

$100(1-\alpha)$ percent confidence interval for μ. Since σ is unknown, we can no longer base our interval on the fact that $\sqrt{n}(\overline{X}-\mu)/\sigma$ is a standard normal random variable. However, by letting $S^2 = \sum_{i=1}^{n}(X_i - \overline{X})^2/(n-1)$ denote the sample variance, then from Corollary 6.5.2 it follows that

$$\sqrt{n}\,\frac{(\overline{X}-\mu)}{S}$$

is a t-random variable with $n-1$ degrees of freedom. Hence, from the symmetry of the t-density function (see Figure 7.2), we have that for any $\alpha \in (0, 1/2)$,

$$P\left\{-t_{\alpha/2,n-1} < \sqrt{n}\,\frac{(\overline{X}-\mu)}{S} < t_{\alpha/2,n-1}\right\} = 1-\alpha$$

or, equivalently,

$$P\{-t_{\alpha/2,n-1}\,\frac{S}{\sqrt{n}} < \bar{X}-\mu < t_{\alpha/2,n-1}\,\frac{S}{\sqrt{n}}\} = 1-\alpha$$

Multiplying all sides of the preceding by -1 and then adding $\bar{X}$ yields that

$$P\left\{\overline{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}} < \mu < \overline{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}\right\} = 1-\alpha$$

Thus, if it is observed that $\overline{X} = \overline{x}$ and $S = s$, then we can say that "with $100(1-\alpha)$ percent confidence"

$$\mu \in \left(\overline{x} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}},\ \overline{x} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}\right)$$

Example 7.3.e. Let us again consider Example 7.3.a but let us now suppose that when the value μ is transmitted at location A then the value received at location B is normal with mean μ and variance σ^2 but with σ^2 being unknown. If 9 successive values are, as in Example 7.3.a, 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, compute a 95 percent confidence interval for μ.

Solution. A simple calculation yields that

$$\overline{x} = 9$$

and

$$s^2 = \frac{\sum x_i^2 - 9(\overline{x})^2}{8} = 9.5$$

or

$$s = 3.082$$

Hence, as $t_{.025,8} = 2.306$, a 95 percent confidence interval for μ is

$$\left[9 - 2.306\frac{(3.082)}{3},\ 9 + 2.306\frac{(3.082)}{3}\right] = (6.63, 11.37)$$

a larger interval than obtained in Example 7.3.a. The reason why the interval just obtained is larger than the one in Example 7.3.a is twofold. The primary reason is that we have a larger estimated variance than in Example 7.3.a. That is, in Example 7.3.a we assumed that σ^2 was known to equal 4, whereas in this example we assumed it to be unknown and our estimate of it turned out to be 9.5, which resulted in a larger confidence interval. In fact, the confidence interval would have been larger than in Example 7.3.a even if our estimate of σ^2 was again 4 because by having to estimate the variance we need to utilize the t-distribution, which has a greater variance and thus a larger spread than the standard normal (which can be used when σ^2 is assumed known). For instance, if it had turned out that $\overline{x} = 9$ and $s^2 = 4$, then our confidence interval would have been

$$(9 - 2.306 \cdot \tfrac{2}{3}, 9 + 2.306 \cdot \tfrac{2}{3}) = (7.46, 10.54)$$

which is larger than that obtained in Example 7.3.a. ■

Remarks

(a) The confidence interval for μ when σ is known is based on the fact that $\sqrt{n}(\overline{X} - \mu)/\sigma$ has a standard normal distribution. When σ is unknown, the foregoing approach is to estimate it by S and then use the fact that $\sqrt{n}(\overline{X} - \mu)/S$ has a t-distribution with $n - 1$ degrees of freedom.

(b) The length of a $100(1-2\alpha)$ percent confidence interval for μ is not always larger when the variance is unknown. For the length of such an interval is $2z_\alpha\sigma/\sqrt{n}$ when σ is known, whereas it is $2t_{\alpha,n-1}S/\sqrt{n}$ when σ is unknown; and it is certainly possible that the sample standard deviation S can turn out to be much smaller than σ. However, it can be shown that the mean length of the interval is longer when σ is unknown. That is, it can be shown that

$$t_{\alpha,n-1}E[S] \geq z_\alpha\sigma$$

Indeed, $E[S]$ is evaluated in Chapter 14 and it is shown, for instance, that

$$E[S] = \begin{cases} .94\,\sigma & \text{when } n = 5 \\ .97\,\sigma & \text{when } n = 9 \end{cases}$$

Since

$$z_{.025} = 1.96, \qquad t_{.025,4} = 2.78, \qquad t_{.025,8} = 2.31$$

the length of a 95 percent confidence interval from a sample of size 5 is $2 \times 1.96\sigma/\sqrt{5} = 1.75\,\sigma$ when σ is known, whereas its expected length is $2 \times 2.78 \times .94\sigma/\sqrt{5} = 2.34\,\sigma$ when σ is unknown — an increase of 33.7 percent. If the sample is of size 9, then the two values to compare are 1.31 σ and 1.49 σ — a gain of 13.7 percent. ■

A one-sided upper confidence interval can be obtained by noting that

$$P\left\{\sqrt{n}\frac{(\overline{X}-\mu)}{S} < t_{\alpha,n-1}\right\} = 1-\alpha$$

or

$$P\left\{\overline{X}-\mu < \frac{S}{\sqrt{n}}t_{\alpha,\,n-1}\right\} = 1-\alpha$$

or

$$P\left\{\mu > \overline{X} - \frac{S}{\sqrt{n}}t_{\alpha,\,n-1}\right\} = 1-\alpha$$

Hence, if it is observed that $\overline{X} = \overline{x}$, $S = s$, then we can assert "with $100(1-\alpha)$ percent confidence" that

$$\mu \in \left(\overline{x} - \frac{s}{\sqrt{n}}t_{\alpha,n-1},\ \infty\right)$$

Similarly, a $100(1-\alpha)$ lower confidence interval would be

$$\mu \in \left(-\infty,\ \overline{x} + \frac{s}{\sqrt{n}} t_{\alpha,\, n-1}\right)$$

We can use R to obtain confidence intervals. To obtain a $100(1-\alpha)$ percent confidence interval for the data values $x_1, ..., x_n$, input

$$> d = c(x_1, \ldots, x_n)$$
$$> l = \text{ mean}(d) - qt(1-\alpha/2, n-1) * \text{sqrt}(\text{var}(d)/n)$$
$$> u = \text{ mean}(d) + qt(1-\alpha/2, n-1) * \text{ sqrt}(\text{var}(d)/n)$$
$$> c(l, u)$$

and press return. For instance, suppose we want to use R to compute a 99 percent confidence interval for the mean of a normal data set when the variance is unknown. Let the data be 5, 9, 12, 8.6, 7.2, 13, 9.5. Then, write the following R commands:

$$> d = c(5, 9, 12, 8.6, 7.2, 13, 9.5)$$
$$> l = \text{ mean}(d) - qt(.995, 6) * \text{sqrt}(\text{var}(d)/7)$$
$$> u = \text{ mean}(d) + qt(.995, 6) * \text{sqrt}(\text{var}(d)/7)$$
$$> c(l, u)$$

Press return and R yields

[1] 5.37346331037313 12.9979652610554

We obtain upper and lower confidence intervals similarly.

Example 7.3.f. Determine a 95 percent confidence interval for the average resting pulse of the members of a health club if a random selection of 15 members of the club yielded the data 54, 63, 58, 72, 49, 92, 70, 73, 69, 104, 48, 66, 80, 64, 77. Also determine a 95 percent lower confidence interval for this mean.

Solution. We use R to obtain the two-sided 95 percent confidence interval, and the 95 percent lower confidence interval as follows:

$$>d = c(54, 63, 58, 72, 49, 92, 70, 73, 69, 104, 48, 66, 80, 64, 77)$$
$$>l = \text{ mean}(d) - qt(.975, 14) * \text{sqrt}(\text{var}(d)/15)$$
$$>u = \text{ mean}(d) + qt(.975, 14) * \text{sqrt}(\text{var}(d)/15)$$
$$>c(l, u)$$

[1] 60.8669366732691, 77.6663966600643

$$>ub = \text{ mean}(d) + qt(.95, 14) * \text{sqrt}(\text{var}(d)/15)$$

[1] 76.16457

Thus, the 95 percent two-sided confidence interval for μ is (60.8669366732691, 77.6663966600643), while the 95 percent lower confidence interval is $(-\infty, 76.16457)$. ■

Our derivations of the $100(1-\alpha)$ percent confidence intervals for the population mean μ have assumed that the population distribution is normal. However, even when this is not the case, if the sample size is reasonably large then the intervals obtained will still be approximate $100(1-\alpha)$ percent confidence intervals for μ. This is true because, by the central limit theorem, $\sqrt{n}(\overline{X}-\mu)/\sigma$ will have approximately a normal distribution, and $\sqrt{n}(\overline{X}-\mu)/S$ will have approximately a t-distribution.

Example 7.3.g. Simulation provides a powerful method for evaluating single and multidimensional integrals. For instance, let f be a function of an r-valued vector $(y_1, \ldots, y_r)$, and suppose that we want to estimate the quantity θ, defined by

$$\theta = \int_0^1 \int_0^1 \cdots \int_0^1 f(y_1, y_2, \ldots, y_r)\, dy_1 dy_2, \ldots, dy_r$$

To accomplish this, note that if $U_1, U_2, \ldots, U_r$ are independent uniform random variables on (0, 1), then

$$\theta = E[f(U_1, U_2, \ldots, U_r)]$$

Now, the values of independent uniform (0, 1) random variables can be approximated on a computer (by so-called *pseudo random numbers*); if we generate a vector of r of them, and evaluate f at this vector, then the value obtained, call it X_1, will be a random variable with mean θ. If we now repeat this process, then we obtain another value, call it X_2, which will have the same distribution as X_1. Continuing on, we can generate a sequence $X_1, X_2, \ldots, X_n$ of independent and identically distributed random variables with mean θ; we then use their observed values to estimate θ. This method of approximating integrals is called *Monte Carlo simulation*.

For instance, suppose we wanted to estimate the one-dimensional integral

$$\theta = \int_0^1 \sqrt{1-y^2}\, dy = E[\sqrt{1-U^2}]$$

where U is a uniform (0, 1) random variable. To do so, let $U_1, \ldots, U_{100}$ be independent uniform (0, 1) random variables, and set

$$X_i = \sqrt{1-U_i^2}, \qquad i = 1, \ldots, 100$$

In this way, we have generated a sample of 100 random variables having mean θ. Suppose that the computer-generated values of $U_1, \ldots, U_{100}$, resulting in

$X_1, \ldots, X_{100}$ having sample mean .786 and sample standard deviation .03. Consequently, since $t_{.025,99} = 1.985$, it follows that a 95 percent confidence interval for θ would be given by

$$.786 \pm 1.985(.003)$$

As a result, we could assert, with 95 percent confidence, that θ (which can be shown to equal $\pi/4$) is between .780 and .792. ■

7.3.2 Prediction intervals

Suppose that $X_1, \ldots, X_n, X_{n+1}$ is a sample from a normal distribution with unknown mean μ and unknown variance σ^2. Suppose further that the values of $X_1, \ldots, X_n$ are to be observed and that we want to use them to predict the value of X_{n+1}. To begin, note that if the mean μ were known, then it would be the natural predictor for X_{n+1}. As it is not known, it seems natural to use its current estimator after observing $X_1, \ldots, X_n$, namely the average of these observed values, as the predicted value of X_{n+1}. That is, we should use the observed value of $\bar{X}_n = \sum_{i=1}^n X_i/n$, as the predicted value of X_{n+1}.

Suppose now that we want to determine an interval in which we predict, with a certain degree of confidence, that X_{n+1} will lie. To obtain such a prediction interval, note that as $\bar{X}_n$ is normal with mean μ and variance σ^2/n, and is independent of X_{n+1} which is normal with mean μ and variance σ^2, it follows that $X_{n+1} - \bar{X}_n$ is normal with mean 0 and variance $sigma^2/n + \sigma^2$. Consequently,

$$\frac{X_{n+1} - \bar{X}_n}{\sigma\sqrt{1 + 1/n}} \qquad \text{is a standard normal random variable.}$$

Because this is independent of $S_n^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2/(n-1)$, it follows from the same argument used to establish Corollary 6.5.2, that replacing σ by its estimator S_n in the preceding expression will yield a t-random variable with $n-1$ degrees of freedom. That is,

$$\frac{X_{n+1} - \bar{X}_n}{S_n\sqrt{1 + 1/n}}$$

is a t-random variable with $n-1$ degrees of freedom. Hence, for any $\alpha \in (0, 1/2)$,

$$P\{-t_{\alpha/2,n-1} < \frac{X_{n+1} - \bar{X}_n}{S_n\sqrt{1 + 1/n}} < t_{\alpha/2,n-1}\} = 1 - \alpha$$

which is equivalent to

$$P\{\bar{X}_n - t_{\alpha/2,n-1}\, S_n\sqrt{1 + 1/n} < X_{n+1} < \bar{X}_n + t_{\alpha/2,n-1}\, S_n\sqrt{1 + 1/n}\}$$

Hence, if the observed values of $\bar{X}_n$ and S_n are, respectively, $\bar{x}_n$ and s_n, then we can predict, with $100(1-\alpha)$ percent confidence, that X_{n+1} will lie between $\bar{x}_n - t_{\alpha/2,n-1}s_n\sqrt{1+1/n}$ and $\bar{x}_n + t_{\alpha/2,n-1}s_n\sqrt{1+1/n}$. That is, with $100(1-\alpha)$ percent confidence, we can predict that

$$X_{n+1} \in \left(\bar{x}_n - t_{\alpha/2,n-1}s_n\sqrt{1+1/n}, \bar{x}_n + t_{\alpha/2,n-1}s_n\sqrt{1+1/n}\right)$$

Example 7.3.h. The following are the number of steps walked in each of the last 7 days

6822 5333 7420 7432 6252 7005 6752

Assuming that the daily number of steps can be thought of as being independent realizations from a normal distribution, give a prediction interval that, with 95 percent confidence, will contain the number of steps that will be walked tomorrow.

Solution. A simple calculation gives that the sample mean and sample variance of the 7 data values are

$$\bar{X}_7 = 6716.57 \qquad S_7 = 733.97$$

Because $t_{.025,6} = 2.447$, and $2.4447 \cdot 733.97\sqrt{1+1/7} = 1920.03$, we can predict, with 95 percent confidence, that tomorrow's number of steps will be between $6716.57 - 1920.03$ and $6716.57 + 1920.03$. That is, with 95 percent confidence, X_8 will lie in the interval (4796.54, 8636.60).

7.3.3 Confidence intervals for the variance of a normal distribution

If $X_1, \ldots, X_n$ is a sample from a normal distribution having unknown parameters μ and σ^2, then we can construct a confidence interval for σ^2 by using the fact that

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$$

Hence,

$$P\left\{\chi^2_{1-\alpha/2,n-1} \le (n-1)\frac{S^2}{\sigma^2} \le \chi^2_{\alpha/2,n-1}\right\} = 1-\alpha$$

or, equivalently,

$$P\left\{\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}} \le \sigma^2 \le \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right\} = 1-\alpha$$

or

$$P\{\mu - 1.645\sigma < \theta < \mu + 1.645\sigma\} = .90$$

it seems reasonable to take

$$1.645\sigma = a \quad \text{or} \quad \sigma = \frac{a}{1.645}$$

Thus, if your prior feelings can indeed be reasonably described by a normal distribution, then that distribution would have mean μ and standard deviation $\sigma = a/1.645$. As a test of whether this distribution indeed fits your prior feelings you might ask yourself such questions as whether you are 95 percent certain that θ will fall between $\mu - 1.96\sigma$ and $\mu + 1.96\sigma$, or whether you are 99 percent certain that θ will fall between $\mu - 2.58\sigma$ and $\mu + 2.58\sigma$, where these intervals are determined by the equalities

$$P\left\{-1.96 < \frac{\theta - \mu}{\sigma} < 1.96\right\} = .95$$
$$P\left\{-2.58 < \frac{\theta - \mu}{\sigma} < 2.58\right\} = .99$$

which hold when θ is normal with mean μ and variance σ^2.

Example 7.8.c. Consider the likelihood function $f(x_1, \ldots, x_n|\theta)$ and suppose that θ is uniformly distributed over some interval (a, b). The posterior density of θ given $X_1, \ldots, X_n$ equals

$$\begin{aligned} f(\theta|x_1, \ldots, x_n) &= \frac{f(x_1, \ldots, x_n|\theta)\, p(\theta)}{\int_a^b f(x_1, \ldots, x_n|\theta)\, p(\theta)\, d\theta} \\ &= \frac{f(x_1, \ldots, x_n|\theta)}{\int_a^b f(x_1, \ldots, x_n|\theta)\, d\theta} \qquad a < \theta < b \end{aligned}$$

Now the *mode* of a density $f(\theta)$ was defined to be that value of θ that maximizes $f(\theta)$. By the foregoing, it follows that the mode of the density $f(\theta|x_1, \ldots, x_n)$ is that value of θ maximizing $f(x_1, \ldots, x_n|\theta)$; that is, it is just the maximum likelihood estimate of θ [when it is constrained to be in (a, b)]. In other words, the maximum likelihood estimate equals the mode of the posterior distribution when a uniform prior distribution is assumed. ■

If, rather than a point estimate, we desire an interval in which θ lies with a specified probability — say $1 - \alpha$ — we can accomplish this by choosing values a and b such that

$$\int_a^b f(\theta|x_1, \ldots, x_n)\, d\theta = 1 - \alpha$$

Example 7.8.d. Suppose that if a signal of value s is sent from location A, then the signal value received at location B is normally distributed with mean s and variance 60. Suppose also that the value of a signal sent at location A is, *a priori*, known to be normally distributed with mean 50 and variance 100. If the value received at location B is equal to 40, determine an interval that will contain the actual value sent with probability .90.

Solution. It follows from Example 7.8.b that the conditional distribution of S, the signal value sent, given that 40 is the value received, is normal with mean and variance given by

$$E[S|\text{data}] = \frac{1/60}{1/60+1/100}40 + \frac{1/100}{1/60+1/100}50 = 43.75$$

$$\text{Var}(S|\text{data}) = \frac{1}{1/60+1/100} = 37.5$$

Hence, given that the value received is 40, $(S-43.75)/\sqrt{37.5}$ has a standard normal distribution and so

$$P\left\{-1.645 < \frac{S-43.75}{\sqrt{37.5}} < 1.645|\text{data}\right\} = .90$$

or

$$P\{43.75 - 1.645\sqrt{37.5} < S < 43.75 + 1.645\sqrt{37.5}|\text{data}\} = .95$$

That is, with *probability* .90, the true signal sent lies within the interval (33.68, 53.82). ■

Problems

1. Let $X_1, \ldots, X_n$ be a sample from the distribution whose density function is

$$f(x) = \begin{cases} e^{-(x-\theta)} & x \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

   Determine the maximum likelihood estimator of θ.
2. Determine the maximum likelihood estimator of θ when $X_1, \ldots, X_n$ is a sample with density function

$$f(x) = \tfrac{1}{2}e^{-|x-\theta|}, \qquad -\infty < x < \infty$$

3. Let $X_1, \ldots, X_n$ be a sample from a normal μ, σ^2 population. Determine the maximum likelihood estimator of σ^2 when μ is known. What is the expected value of this estimator?

4. Determine the maximum likelihood estimates of a and λ when $X_1, \ldots, X_n$ is a sample from the Pareto density function

$$f(x) = \begin{cases} \lambda a^{\lambda} x^{-(\lambda+1)}, & \text{if } x \geq a \\ 0, & \text{if } x < a \end{cases}$$

5. Suppose that $X_1, \ldots, X_n$ are normal with mean μ_1; $Y_1, \ldots, Y_n$ are normal with mean μ_2; and $W_1, \ldots, W_n$ are normal with mean $\mu_1 + \mu_2$. Assuming that all $3n$ random variables are independent with a common variance, find the maximum likelihood estimators of μ_1 and μ_2.
6. River floods are often measured by their discharges (in units of feet cubed per second). The value v is said to be the value of a 100-year flood if

$$P\{D \geq v\} = .01$$

where D is the discharge of the largest flood in a randomly chosen year. The following table gives the flood discharges of the largest floods of the Blackstone River in Woonsocket, Rhode Island, in each of the years from 1929 to 1965. Assuming that these discharges follow a lognormal distribution, estimate the value of a 100-year flood.

Table 7.4 Annual Floods of the Blackstone River (1929–1965).

Year	Flood Discharge (ft^3/s)
1929	4,570
1930	1,970
1931	8,220
1932	4,530
1933	5,780
1934	6,560
1935	7,500
1936	15,000
1937	6,340
1938	15,100
1939	3,840
1940	5,860
1941	4,480
1942	5,330
1943	5,310
1944	3,830
1945	3,410

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Table 7.4 (*continued*)

Year	Flood Discharge (ft^3/s)
1946	3830
1947	3150
1948	5810
1949	2030
1950	3620
1951	4920
1952	4090
1953	5570
1954	9400
1955	32,900
1956	8710
1957	3850
1958	4970
1959	5398
1960	4780
1961	4020
1962	5790
1963	4510
1964	5520
1965	5300

7. Recall that X is said to have a lognormal distribution with parameters μ and σ^2 if $\log(X)$ is normal with mean μ and variance σ^2. Suppose X is such a lognormal random variable.
   a. Find $E[X]$.
   b. Find $\text{Var}(X)$.
   Hint: Make use of the formula for the moment generating function of a normal random variable.
   c. The following are, in minutes, travel times to work over a sequence of 10 days.

   42, 28, 53, 57, 67, 39, 35, 50, 44, 39

   Assuming an underlying lognormal distribution, use the data to estimate the mean travel time.
8. An electric scale gives a reading equal to the true weight plus a random error that is normally distributed with mean 0 and standard deviation $\sigma = .1$ mg. Suppose that the results of five successive weighings of the same object are as follows: 3.142, 3.163, 3.155, 3.150, 3.141.

a. Determine a 95 percent confidence interval estimate of the true weight.
b. Determine a 99 percent confidence interval estimate of the true weight.

9. The PCB concentration of a fish caught in Lake Michigan was measured by a technique that is known to result in an error of measurement that is normally distributed with a standard deviation of .08 ppm (parts per million). Suppose the results of 10 independent measurements of this fish are

11.2, 12.4, 10.8, 11.6, 12.5, 10.1, 11.0, 12.2, 12.4, 10.6

a. Give a 95 percent confidence interval for the PCB level of this fish.
b. Give a 95 percent lower confidence interval.
c. Give a 95 percent upper confidence interval.

10. The standard deviation of test scores on a certain achievement test is 11.3. If a random sample of 81 students had a sample mean score of 74.6, find a 90 percent confidence interval estimate for the average score of all students.

11. Let $X_1, \ldots, X_n, X_{n+1}$ be a sample from a normal population having an unknown mean μ and variance 1. Let $\bar{X}_n = \sum_{i=1}^{n} X_i/n$ be the average of the first n of them.
a. What is the distribution of $X_{n+1} - \bar{X}_n$?
b. If $\bar{X}_n = 4$, give an interval that, with 90 percent confidence, will contain the value of X_{n+1}.

12. If $X_1, \ldots, X_n$ is a sample from a normal population whose mean μ is unknown but whose variance σ^2 is known, show that $(-\infty, \overline{X} + z_\alpha \sigma/\sqrt{n})$ is a $100(1-\alpha)$ percent lower confidence interval for μ.

13. A sample of 20 cigarettes is tested to determine nicotine content and the average value observed was 1.2 mg. Compute a 99 percent two-sided confidence interval for the mean nicotine content of a cigarette if it is known that the standard deviation of a cigarette's nicotine content is $\sigma =$ .2 mg.

14. In Problem 13, suppose that the population variance is not known in advance of the experiment. If the sample variance is .04, compute a 99 percent two-sided confidence interval for the mean nicotine content.

15. In Problem 14, compute a value c for which we can assert "with 99 percent confidence" that c is larger than the mean nicotine content of a cigarette.

16. Suppose that when sampling from a normal population having an unknown mean μ and unknown variance σ^2, we wish to determine a sample size n so as to guarantee that the resulting $100(1-\alpha)$ percent confidence interval for μ will be of size no greater than A, for given values α and A. Explain how we can approximately do this by a double sampling scheme that first takes a subsample of size 30 and then chooses the total sample size by using the results of the first subsample.

17. The following data resulted from 24 independent measurements of the melting point of lead.

330°C	322°C	345°C
328.6°C	331°C	342°C
342.4°C	340.4°C	329.7°C
334°C	326.5°C	325.8°C
337.5°C	327.3°C	322.6°C
341°C	340°C	333°C
343.3°C	331°C	341°C
329.5°C	332.3°C	340°C

Assuming that the measurements can be regarded as constituting a normal sample whose mean is the true melting point of lead, determine a 95 percent two-sided confidence interval for this value. Also determine a 99 percent two-sided confidence interval.

18. The following are scores on IQ tests of a random sample of 18 students at a large eastern university.

130, 122, 119, 142, 136, 127, 120, 152, 141,
132, 127, 118, 150, 141, 133, 137, 129, 142

a. Construct a 95 percent confidence interval estimate of the average IQ score of all students at the university.
b. Construct a 95 percent lower confidence interval estimate.
c. Construct a 95 percent upper confidence interval estimate.

19. Suppose that a random sample of nine recently sold houses in a certain city resulted in a sample mean price of $222,000, with a sample standard deviation of $22,000. Give a 95 percent upper confidence interval for the mean price of all recently sold houses in this city.

20. A company self-insures its large fleet of cars against collisions. To determine its mean repair cost per collision, it has randomly chosen a sample of 16 accidents. If the average repair cost in these accidents is $2200 with a sample standard deviation of $800, find a 90 percent confidence interval estimate of the mean cost per collision.

21. A standardized test is given annually to all sixth-grade students in the state of Washington. To determine the average score of students in her district, a school supervisor selects a random sample of 100 students. If the sample mean of these students' scores is 320 and the sample standard deviation is 16, give a 95 percent confidence interval estimate of the average score of students in that supervisor's district.

22. Each of 20 science students independently measured the melting point of lead. The sample mean and sample standard deviation of these measurements were (in degrees centigrade) 330.2 and 15.4, respectively. Construct **(a)** a 95 percent and **(b)** a 99 percent confidence interval estimate of the true melting point of lead.

23. A random sample of 300 CitiBank VISA cardholder accounts indicated a sample mean debt of \$1220 with a sample standard deviation of \$840. Construct a 95 percent confidence interval estimate of the average debt of all cardholders.
24. In Problem 23, find the smallest value v that "with 90 percent confidence," exceeds the average debt per cardholder.
25. Verify the formula given in Table 7.1 for the $100(1-\alpha)$ percent lower confidence interval for μ when σ is unknown.
26. The following are the daily number of steps taken by a certain individual in 20 weekdays.

2,100	1,984	2,072	1,898
1,950	1,992	2,096	2,103
2,043	2,218	2,244	2,206
2,210	2,152	1,962	2,007
2,018	2,106	1,938	1,956

Assuming that the daily number of steps is normally distributed, construct **(a)** a 95 percent and **(b)** a 99 percent two-sided confidence interval for the mean number of steps. **(c)** Determine the largest value v that, "with 95 percent confidence," will be less than the mean range.
27. Studies were conducted in Los Angeles to determine the carbon monoxide concentration near freeways. The basic technique used was to capture air samples in special bags and to then determine the carbon monoxide concentration by using a spectrophotometer. The measurements in ppm (parts per million) over a sampled period during the year were 102.2, 98.4, 104.1, 101, 102.2, 100.4, 98.6, 88.2, 78.8, 83, 84.7, 94.8, 105.1, 106.2, 111.2, 108.3, 105.2, 103.2, 99, 98.8. Compute a 95 percent two-sided confidence interval for the mean carbon monoxide concentration.
28. A set of 10 determinations, by a method devised by the chemist Karl Fischer, of the percentage of water in a methanol solution yielded the following data.

.50, .55, .53, .56, .54,
.57, .52, .60, .55, .58

Assuming normality, use these data to give a 95 percent confidence interval for the actual percentage.
29. Suppose that $U_1, U_2, \ldots$ is a sequence of independent uniform (0,1) random variables, and define N by

$$N = \min\{n : U_1 + \cdots + U_n > 1\}$$

That is, N is the number of uniform (0, 1) random variables that need to be summed to exceed 1. Use random numbers to determine the value of 36 random variables having the same distribution as N, and then use

these data to obtain a 95 percent confidence interval estimate of $E[N]$. Based on this interval, guess the exact value of $E[N]$.

30. An important issue for a retailer is to decide when to reorder stock from a supplier. A common policy used to make the decision is of a type called s, S: The retailer orders at the end of a period if the on-hand stock is less than s, and orders enough to bring the stock up to S. The appropriate values of s and S depend on different cost parameters, such as inventory holding costs and the profit per item sold, as well as the distribution of the demand during a period. Consequently, it is important for the retailer to collect data relating to the parameters of the demand distribution. Suppose that the following data give the numbers of a certain type of item sold in each of 30 weeks.

    14, 8, 12, 9, 5, 22, 15, 12, 16, 7, 10, 9, 15, 15, 12,
    9, 11, 16, 8, 7, 15, 13, 9, 5, 18, 14, 10, 13, 7, 11

    Assuming that the numbers sold each week are independent random variables from a common distribution, use the data to obtain a 95 percent confidence interval for the mean number sold in a week.

31. A random sample of 16 professors at a large private university yielded a sample mean annual salary of \$90,450 with a sample standard deviation of \$9400. Determine a 95 percent confidence interval of the average salary of all professors at that university.

32. Let $X_1, \ldots, X_{n+1}$ be a sample from a population with mean μ and variance σ^2. As noted in the text, the natural predictor of X_{n+1} based on the data values $X_1, \ldots, X_n$ is $\bar{X}_n = \sum_{i=1}^{n} X_i/n$. Determine the mean square error of this predictor. That is, find $E[(X_{n+1} - \bar{X}_n)^2]$.

33. National Safety Council data show that the number of accidental deaths due to drowning in the United States in the years from 1990 to 1993 were (in units of one thousand) 5.2, 4.6, 4.3, 4.8. Use these data to give an interval that will, with 95 percent confidence, contain the number of such deaths in 1994.

34. The daily dissolved oxygen concentration for a water stream has been recorded over 30 days. If the sample average of the 30 values is 2.5 mg/liter and the sample standard deviation is 2.12 mg/liter, determine a value which, with 90 percent confidence, exceeds the mean daily concentration.

35. Verify the formulas given in Table 7.1 for the $100(1 - \alpha)$ percent lower and upper confidence intervals for σ^2.

36. The capacities (in ampere-hours) of 10 batteries were recorded as follows:

    140, 136, 150, 144, 148, 152, 138, 141, 143, 151

    a. Estimate the population variance σ^2.
    b. Compute a 99 percent two-sided confidence interval for σ^2.